Dr. Calum Murray

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Academic positions

September 2024 - Ongoing — **Postdoctoral researcher**
CosmoStat, CEA Saclay
Advisor: Dr. Martin Kilbinger & Dr. Samuel Farrens

September 2022 - September 2024 — **Postdoctoral researcher**
AstroParticle and Cosmology laboratory, Université de Paris
Advisor: Prof. James G. Bartlett

November 2020 - September 2022 — **Postdoctoral researcher**
Laboratory of Subatomic Physics and Cosmology, Université Grenoble Alpes
Advisor: Dr. Céline Combet

Education

October 2017 - October 2020 — **PhD, Université Paris Cité**

- Thesis title: Galaxy cluster cosmology with the Euclid Dark Energy Survey
- Advisor: Prof. James G. Bartlett

September 2012 - May 2017 — **Master of Mathematical Physics (Mphys), University of Edinburgh**

- Thesis title: The colours of galaxies
- Advisor: Prof. Robert Mann

Collaborations

October 2024 - Present — **The Ultraviolet Near Infrared Optical Northern Survey**

- Measuring the intrinsic alignment of large-red galaxies using UNIONS galaxy shape measurements.
- Contributing to the cosmic shear analysis: checking how different galaxy sample selections change the shear auto-correlation measurements and the correlation of the shear signal with different survey systematic effects.

October 2020 - Present — **Large Synoptic Survey Telescope Dark Energy Science Collaboration (LSST DESC)**

- Added the computation of the two-halo term (the lensing signal from structures correlated to the cluster but not within it) to the DESC cluster weak lensing processing function CLMM.
- Validated the computation of the weak lensing signal around elliptical halos for the DESC cluster weak lensing processing function CLMM through comparision with my own code.

October 2017 - Present — **Euclid Consortium**

- Assessing selection effects for cluster cosmology by processing the entire 5,150 square degrees of the Euclid Flagship simulation with the Euclid cluster pipeline.
- Co-lead of the Euclid Galaxy Clusters work package on the selection and definition of the Euclid galaxy cluster catalogue.
- Member of the Euclid processing function COMB-CL, which produces weak lensing information for galaxy clusters. I have improved the code speed for large datasets by a factor of 50.
- Produced the first Euclid stacked weak lensing shear profiles around Euclid detected galaxy clusters.
- Produced colour images for each of the detected galaxy clusters within the Euclid quick-release, in order to provide visual verification of the clusters and a lack of obvious artifacts (such as stellar contamination).
- Validating the performance of the Euclid galaxy catalogue in crowded fields using source injection, within the Euclid Science ground segment, who are responsible for producing the Euclid galaxy catalogue.

Teaching and Mentorship

Lecturing and Course Development

August 2025	Lecturer, Euclid Summer School. I will deliver a 6-hour course on the observational aspects of weak gravitational lensing.
June 2025	Lecturer, COLOURS Workshop. Developed and taught a workshop on analysing Euclid's first data-release.

Student Supervision

May - June 2019	Margaux Scavini, Undergraduate student project on strong gravitational lensing for massive neutrino and massive photon mass measurements
April - June 2022	Mahmoud Osman, Masters student project on reducing the scatter of optical mass proxies for galaxy clusters
May - July 2023	Théo Courty, Masters student project on intrinsic alignments of galaxies
April - September 2024	Huy Nguyen Phu, Masters student project on galactic acceleration using quasar proper motions (article in preparation)
January - September 2025	Antonin Corinaldi, Masters student project on intrinsic alignment with UNIONS and DESI (article in preparation), co-supervised with Dr. Martin Kilbinger

Presentations

Selected invited seminars

- Le comptage d'amas de galaxies dans notre Univers avec Euclid
  27ème Congrès Général de la Société Française de Physique, Troyes, France, July 2025
- Constraining f (R) gravity with cross-correlation of galaxies and cosmic microwave background lensing
  Max Planck Institute for Extraterrestrial Physics, Munich, Germany, January 2025
- Beyond cosmic variance: galaxy cluster weak lensing
  Royal Observatory of Edinburgh, Edinburgh, United Kingdom , April 2024
- Robust estimation of the quasar dipole
  Laboratoire Univers et Particules de Montpellier, Montpellier, France , March 2024
- Gravitational lensing by matter currents
  CPPM, Marseille, France, February 2023
- Individual weak lensing masses with HSC
  LPSC, Grenoble, France, January 2021
- Individual weak lensing masses with HSC
  CosmoStat, CEA Saclay, France , January 2020
- Individual weak lensing masses for Euclid
  Observatoire de Paris, Meudon, France , December 2018

In addition I have presented work my research at 8 international conferences (including Rencontres de Moriond, Tracing Cosmic Evolution with Clusters of Galaxies Sexten, Progress on Old and New Themes in cosmology Avignon, and many other conferences), 25 Euclid collaboration meetings (including Euclid France, the general Euclid Consortium meeting, the Euclid galaxy clusters science working group, the Euclid theory science working group), and 8 Dark Energy Science Collaboration meetings.

Service and Public Outreach

January 2025	Peer reviewer for the American Astronomical Society (AAS) journals.
November 2023 - Ongoing	Co-organiser for Astronomy on Tap, Paris. To date, I have organised 12 talks for a general audience, held in an informal bar setting to make astronomy accessible and engaging with the general public.
March 2022 - October 2024	APC cluster club, I organised the weekly journal club for researchers at the APC interested in galaxy clusters.
November 2022 and December 2024	Rencontres Ciel et de l'Espace; a conference for the general public at the Cité de la science et de l'industrie in Paris. Twice I have volunteered at a stand, for Euclid and Rubin, to explain the current activities of each of these cosmogical surveys.
November 2021	Euclid France, LPSC Grenoble, Local Organising Committee
February 2018 - October 2020	Cosmo Journal Club, I organised the biweekly journal club for the APC cosmology group
November 2019	Recontres des Jeunes Physicien.ne.s (RJP), College de France, Paris; I contributed to the organisation of the RJP, which brings together physics PhD students in Paris.

Publications

Peer reviewed publications (with significant contributions)

- Euclid collaboration: S. Bhargava, C. Benoist, A. H. Gonzalez, ..., **C. Murray**, et al. 'Euclid Quick Data Release (Q1). First detections from the galaxy cluster workflow.' *arXiv preprint arXiv:2503.19196* (2025).
- **C. Murray**, C. Combet, C. Payerne, M. Ricci 'Filtering out large-scale noise for cluster weak lensing mass estimation', Astronomy & Astrophysics , 697, A141 (2025)
- C. Payerne, **C. Murray**, C. Combet, C. Doux, M Penna-Lima. 'Cluster abundance cosmology: Including Super-Sample Covariance in the unbinned likelihood' https://arxiv.org/abs/2401.10024
- R. Kou, **C. Murray**, J.G. Bartlett 'Constraining f (R) gravity with cross-correlation of galaxies and cosmic microwave background lensing' https://arxiv.org/abs/2311.09936
- C. Payerne, **C. Murray**, C. Combet, C. Doux, A. Fumagalli, M Penna-Lima (2023). 'Testing the accuracy of likelihoods for cluster abundance cosmology.' Monthly Notices of the Royal Astronomical Society, 520(4), 6223-6236.

- **C. Murray** 'The effects of lensing by local structures on the dipole of radio source counts.' Monthly Notices of the Royal Astronomical Society 510.2 (2022): 3098-3101.
- **C. Murray**, J.G. Bartlett, E. Artis, J.B. Melin 'Measuring weak lensing masses on individual clusters.' Monthly Notices of the Royal Astronomical Society 512.4 (2022): 4785-4791
- E. Artis, J.B. Melin, J.G. Bartlett and **C. Murray** 'Impact of the calibration of the halo mass function on galaxy cluster number count cosmology.' Astronomy & Astrophysics 649 (2021): A47.

In preparation

- **C. Murray**, J.G. Bartlett, R. Kou 'Gravitational lensing by matter currents', here are slides on this research presented at an LSST France meeting.
- Euclid collaboration: **C. Murray**, S. Bhargava, et al. 'Euclid preparation: covariances between the observed properties of galaxy clusters with the Euclid Flagship simulation'
- **C. Murray**, M. Kilbinger 'Tidal torque and the orientations of luminous red galaxies with UNIONS and DESI DR1'

In addition I am a coauthor of 8 published Euclid papers, 1 UNIONS paper recently submitted for publication, and 2 DESC papers under internal review.

References

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